Operation Management

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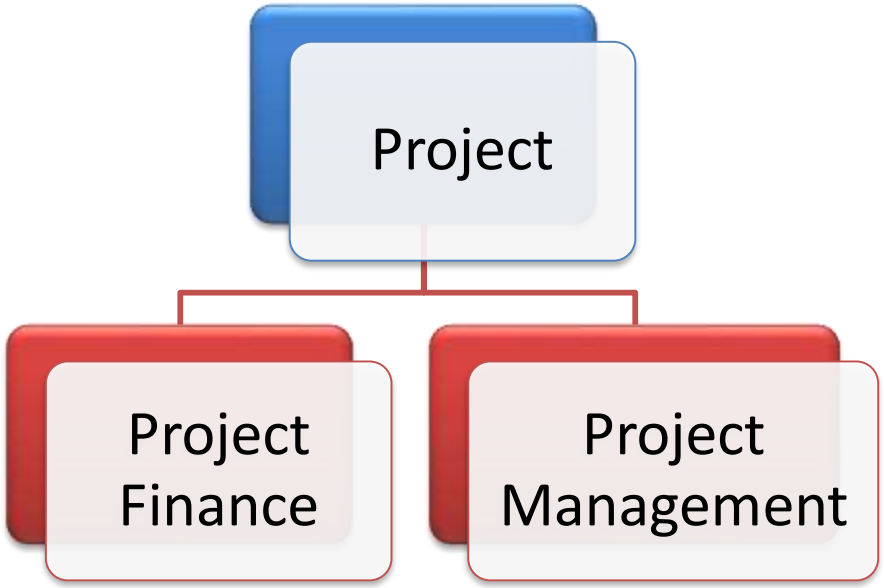
<MBAZG526/QMZG526, Operation Management>
Lecture No. 15 & 16

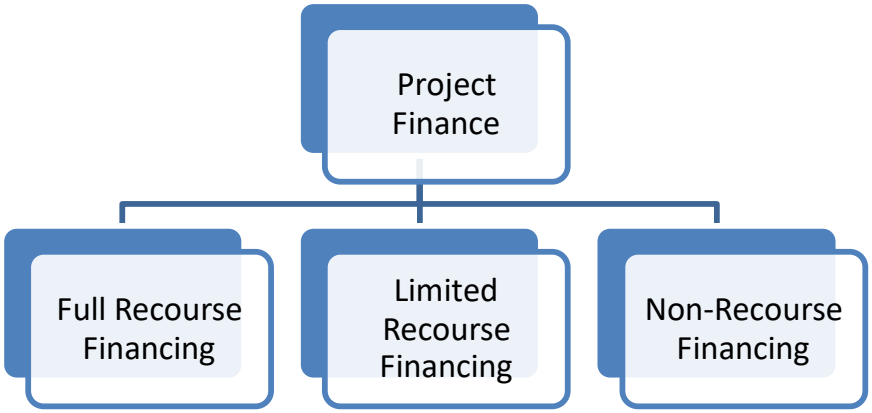
Project



Project Finance deals with large capital investment (debt instruments).

Project Management deals with Project Management Principles including Project Scheduling, Planning and Control.

Project Finance



- Always Choose Non-Recourse Financing.
- Projects is assessed on Stand-alone basis.
- Ring-fencing.
- To prevent negative impact of high leverage on fund raising activities.
- Project Level Risk only.

Project

Project

Project Management

Project

A project is a temporary endeavor undertaken to create a unique product, service, or result.

PMBOK 7th

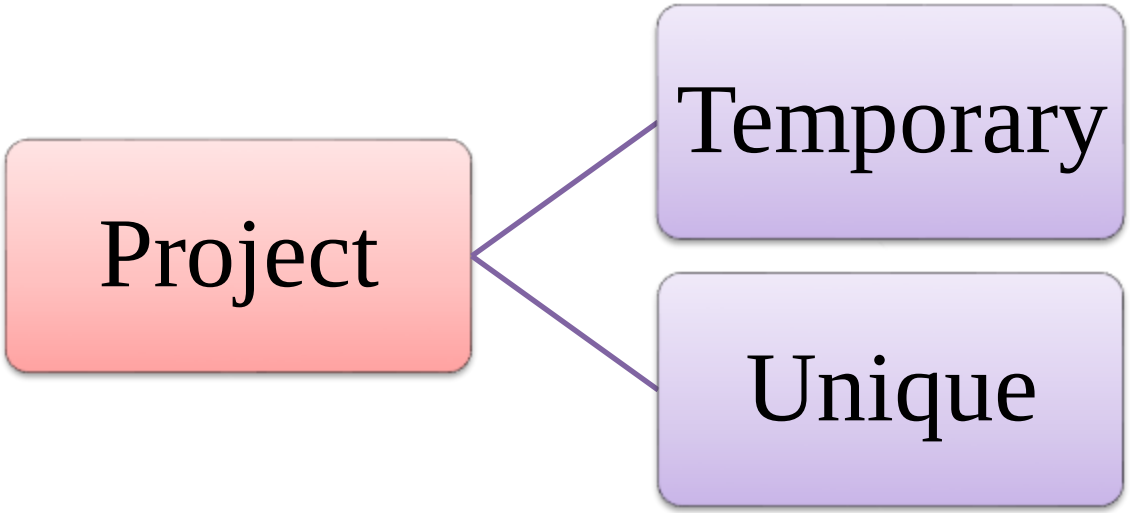
Non- Project

Activities that are ongoing and do not stop when the project completes.

Project Management

Quasi Project

Endeavor with no specific task, no specific budget and no specific deadlines.



Project

- Temporary does not mean short duration.
- Temporary means definite beginning and end.
- Projects end when :

a) Objectives are met or will not met.

b) Funds are depleted.

c) Need no longer exists

d) Resources are no longer available.

e) Legalities or convenience terminates the project.

Project

- Unique means resulting in change from current state to future state.
- May or may not have transition stage in a project.

Project Management

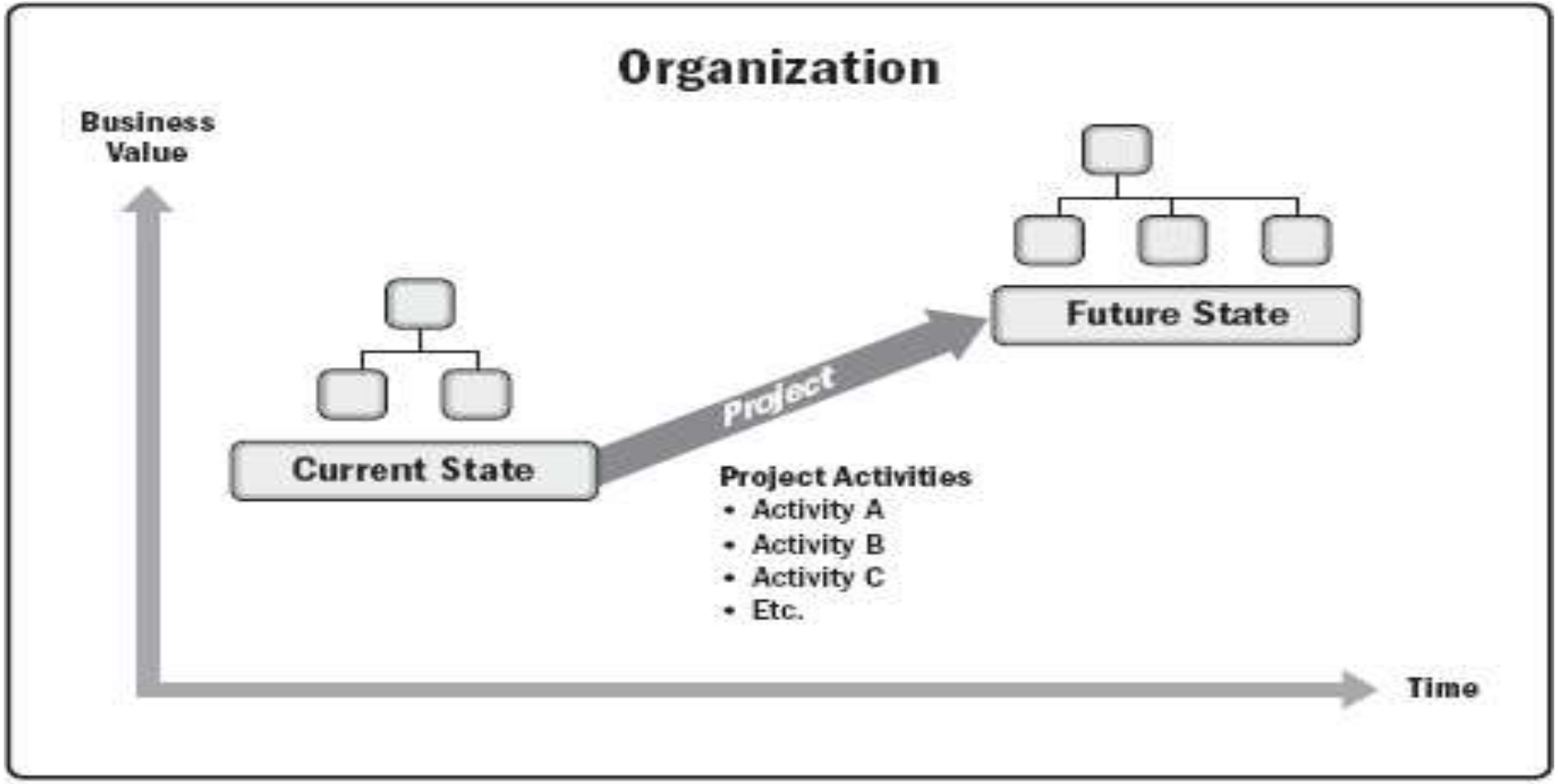
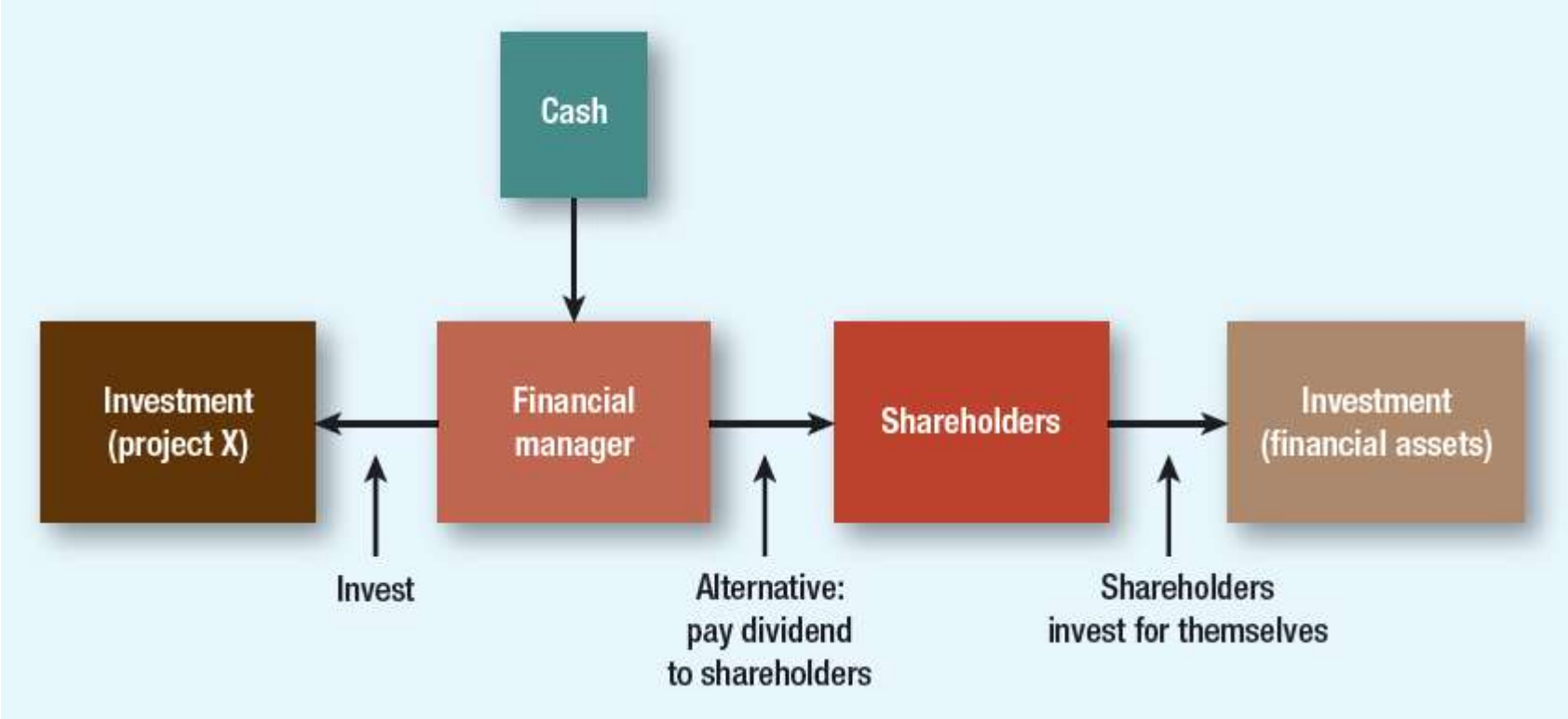


Figure 1-1. Organizational State Transition via a Project

Source: PMBOK

Why Project?


Cash
Investment (project X)
Financial manager
Shareholders
Investment (financial assets)
Invest
Alternative: pay dividend to shareholders
Shareholders invest for themselves

Why Project?

Tangible Business Value	Intangible Business Value
Monetary assets	Goodwill and Reputation
Stakeholder equity	Brand Recognition
Fixtures and tools	Public Benefit
Market Share	Trademarks
	Strategic alignment

How do We Select Project

- Sacred Cow.
- Operating necessity.
- Competitive necessity.
- Product Line Extension.
- Comparative Benefit.
- Sustainability.

Project Feasibility Analysis

Feasibility Analysis

Market Feasibility.

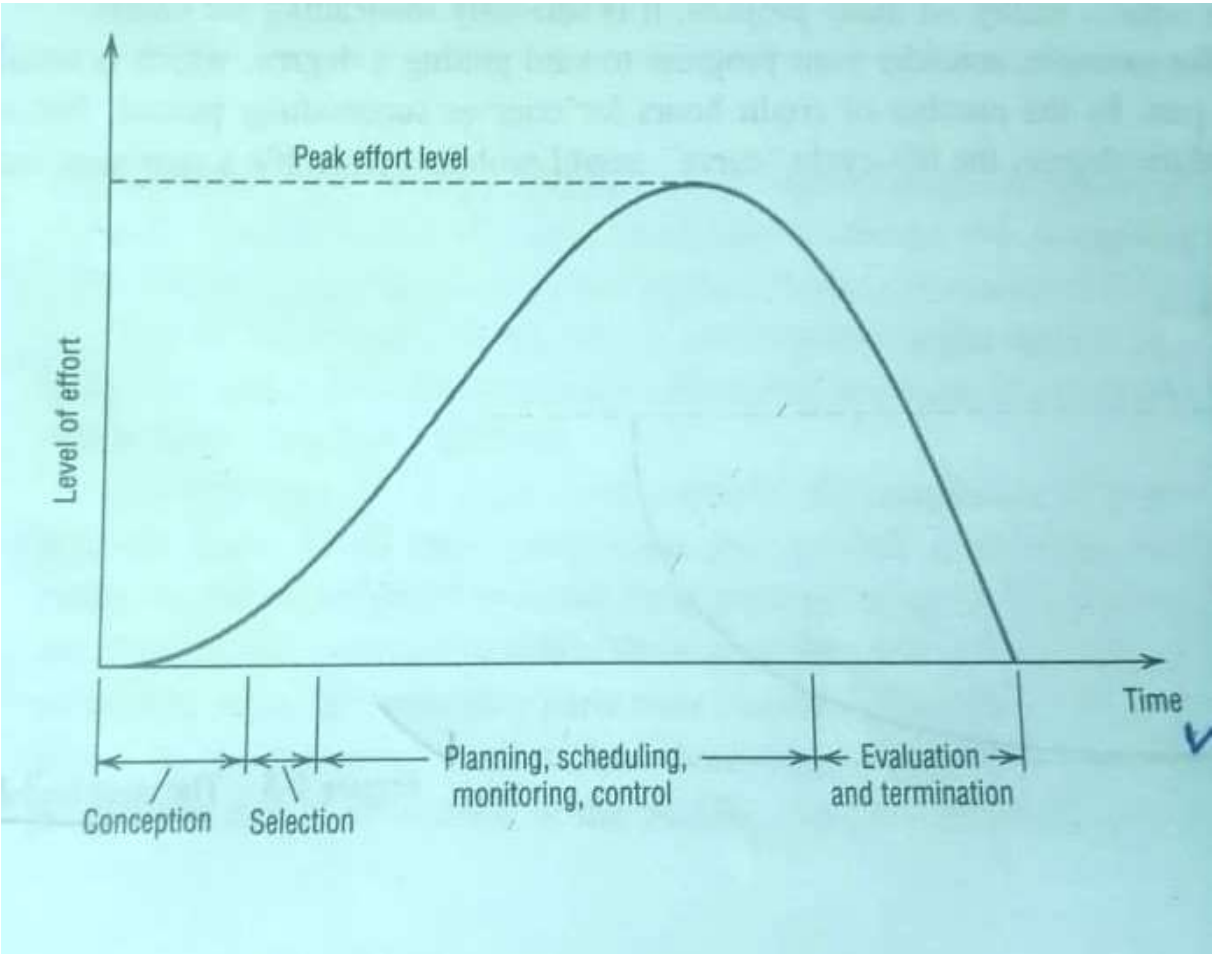
Technical Feasibility.

Financial Feasibility.

Environmental and Social Feasibility.

Project Lifecycle

1. Conception.
2. Selection.
3. Planning.
4. Scheduling.
5. Monitoring.
6. Control
7. Evaluation.
8. Termination



Source: Meredith, Mantel, Shafer

Phases of Project Management

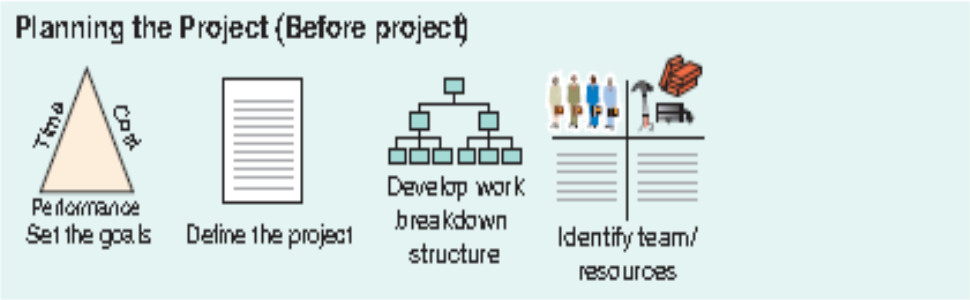
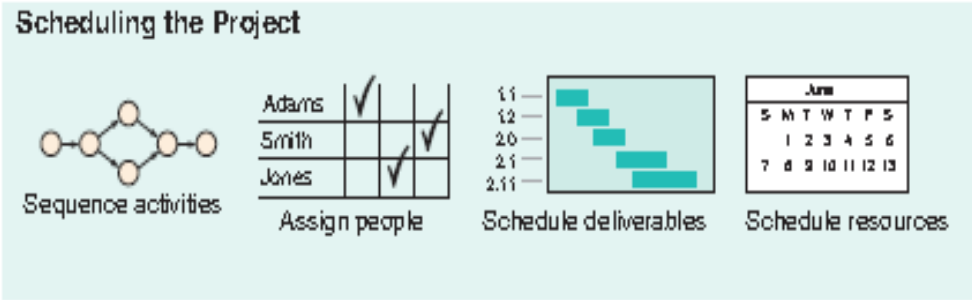


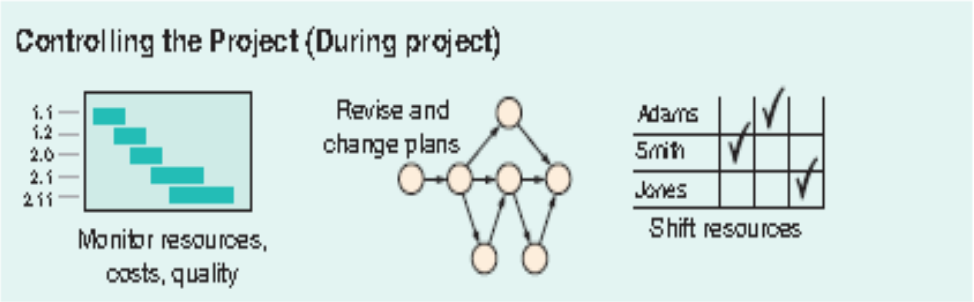
Figure 3.1

Project Planning, Scheduling, and Controlling



STUDENT TIP

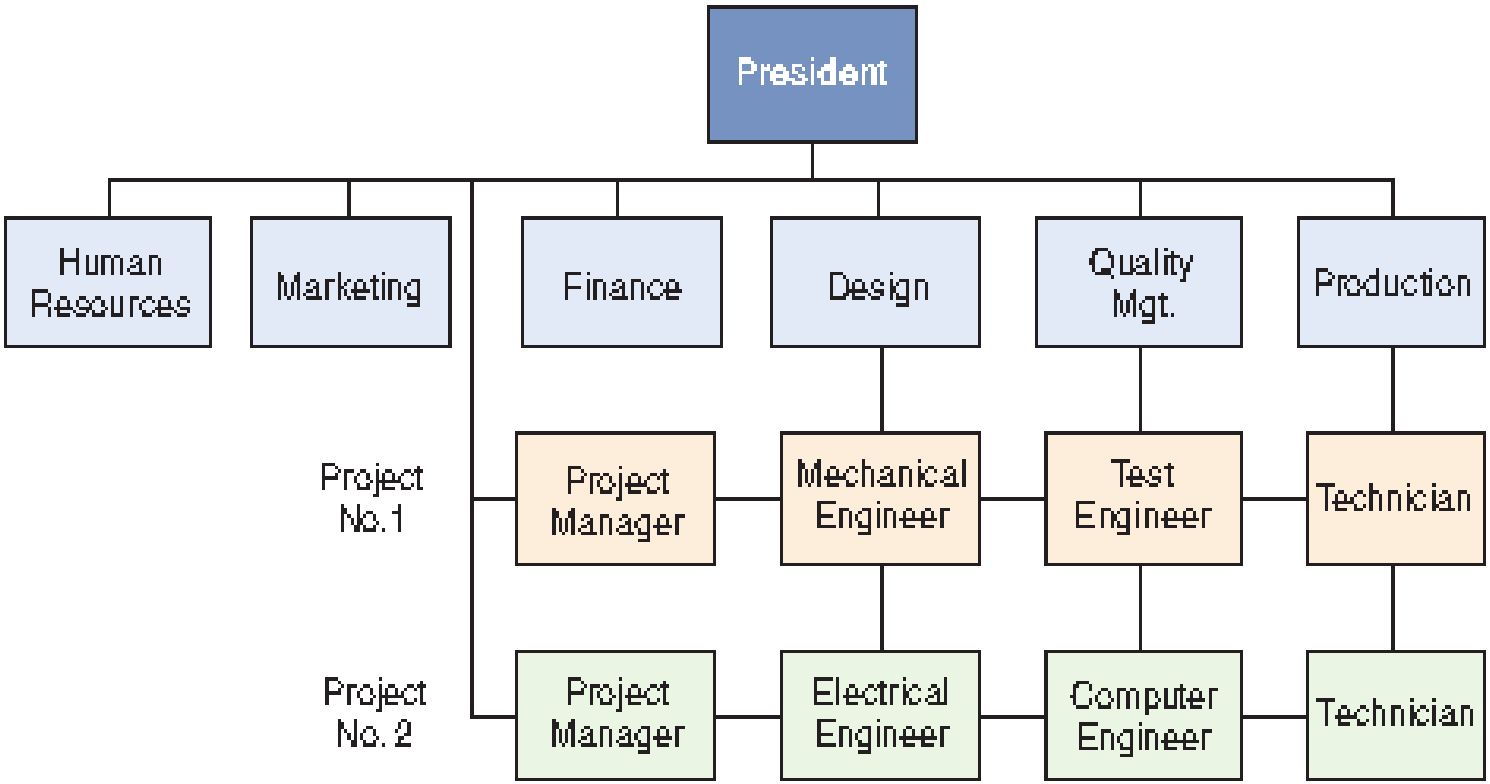
Managers must "make the plan and then work the plan."



Source: Heizer, Render, Muckson

Any Question

Project Organization



Source: Heizer, Render, Muckson

Project Manager

- Any person who can manage the things, lead people, who can get the things done by working in a system and framework can be the project manager.
- The project manager should be an active communicator (listening and speaking).
- The project manager should be extrovert.
- The project manager should be diplomatic.

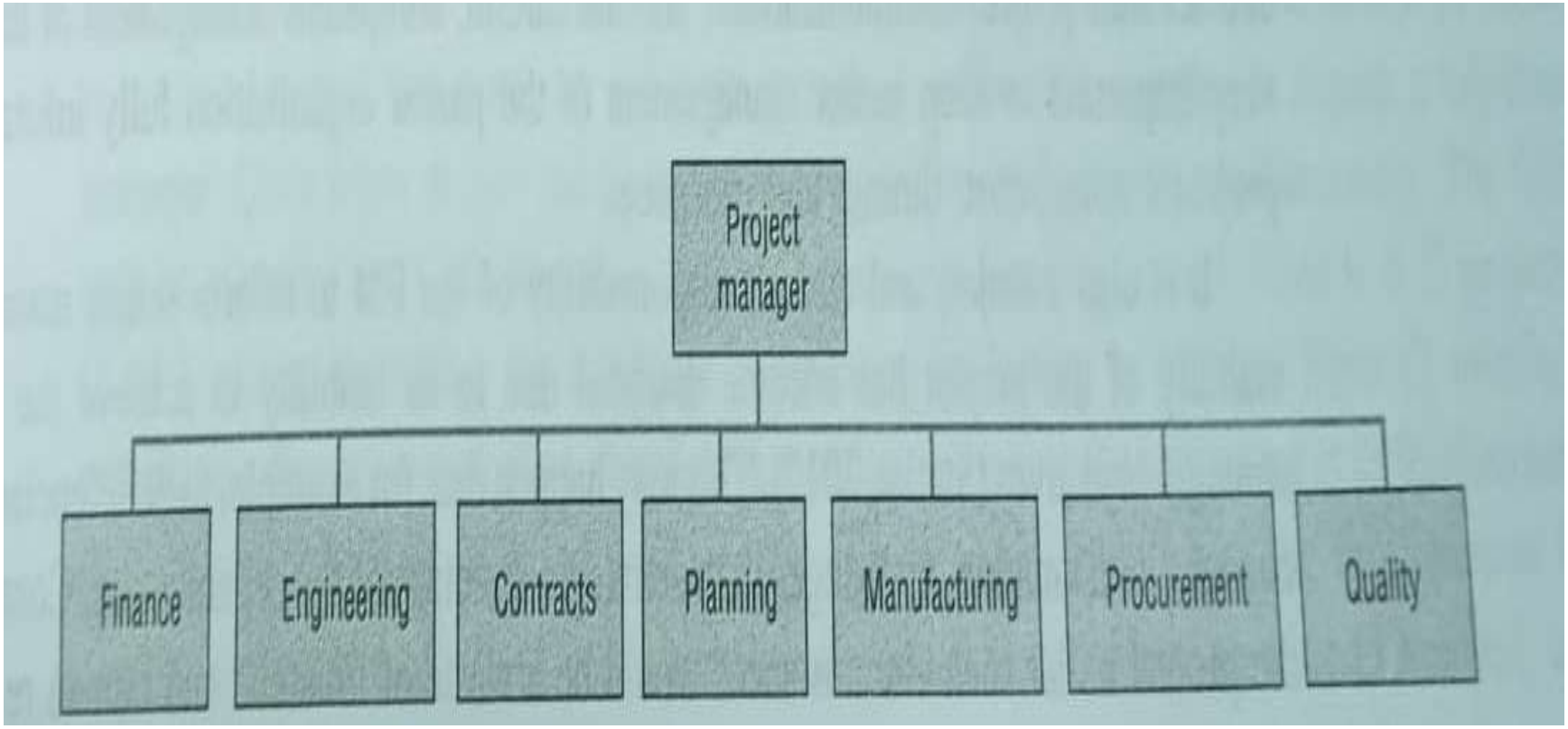
Project Manager

- The project manager need to be chosen at the start of the project.
- The project manager begins with the budget, schedule and resources needed.
- Deals with Scope Creep.
- Making Project Goal Trade-off.

Attributes of Project Manager

- Available.
- Mature.
- Strong-head but polite.
- Good terms with all.
- Someone who can keep the project team happy.
- Worked in several departments.
- Can walk on the waters.

Project Manager



Source : PMBOK

Project Manager

- Project manager are usually generalists.
- It would be very unusual for a project manager to have all the technical skills that are used in their projects.
- Project managers lack the technical skills to evaluate much of the work performed on a particular project.
- Maintaining a balanced outlook.
- Negotiation.
- Deal with Power and Politics.

Work Breakdown Structure (WBS)

- Decomposition of the project Scope.
- Subdivide the project work.
- Smallest item is work package.
- Deliverables oriented.
- Major component of the scope baseline.
- Deterrent to scope changes.

Work Breakdown Structure (WBS)

- Some activities must be done subsequently.
- Some activities may be done simultaneously.
- Many things must happen when and how they are supposed to happen.
- Each detail is uncertain and subjected to risk.

WBS

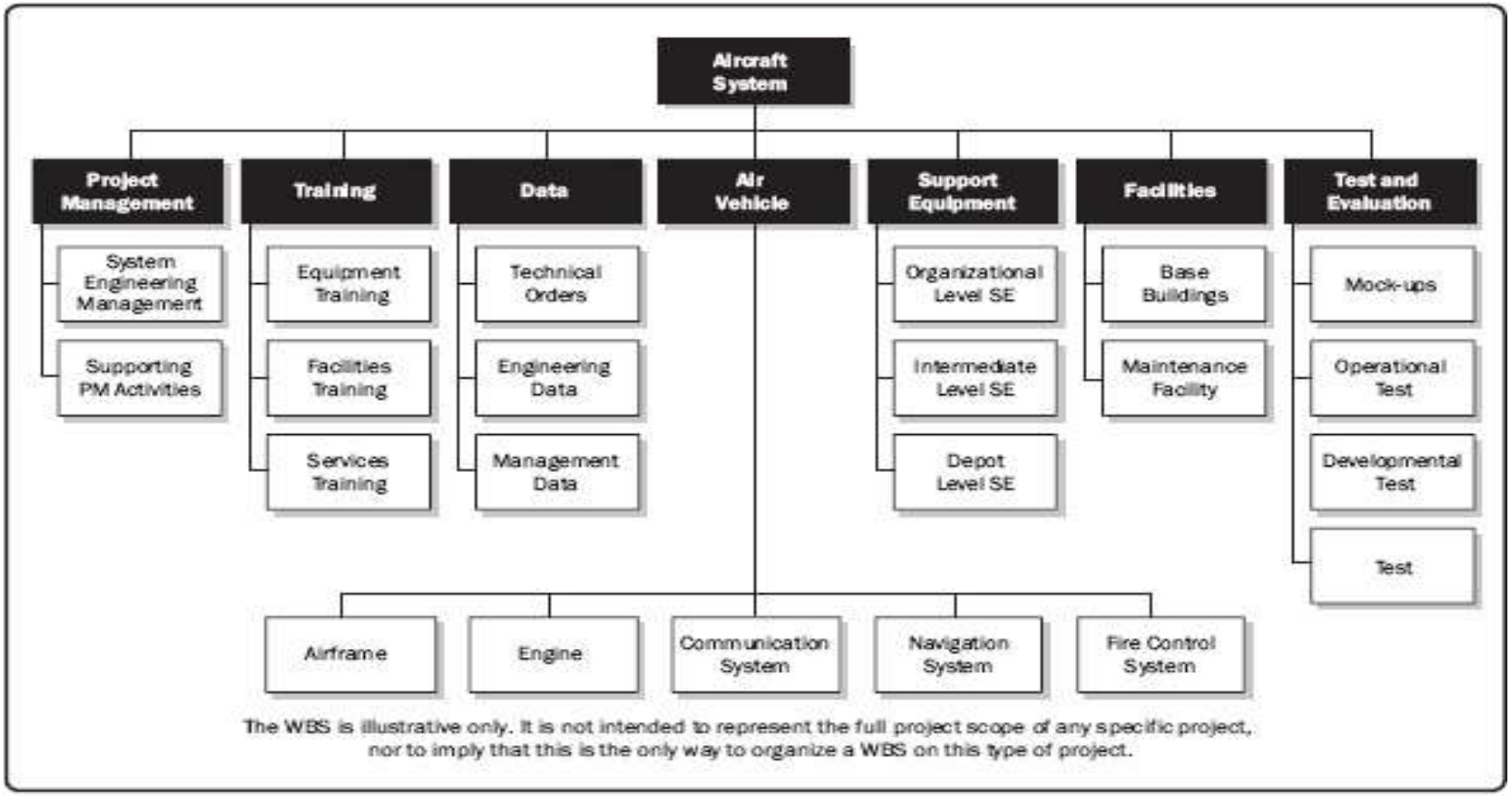


Figure 5-14. Sample WBS with Major Deliverables

Source : PMBOK

Any Question

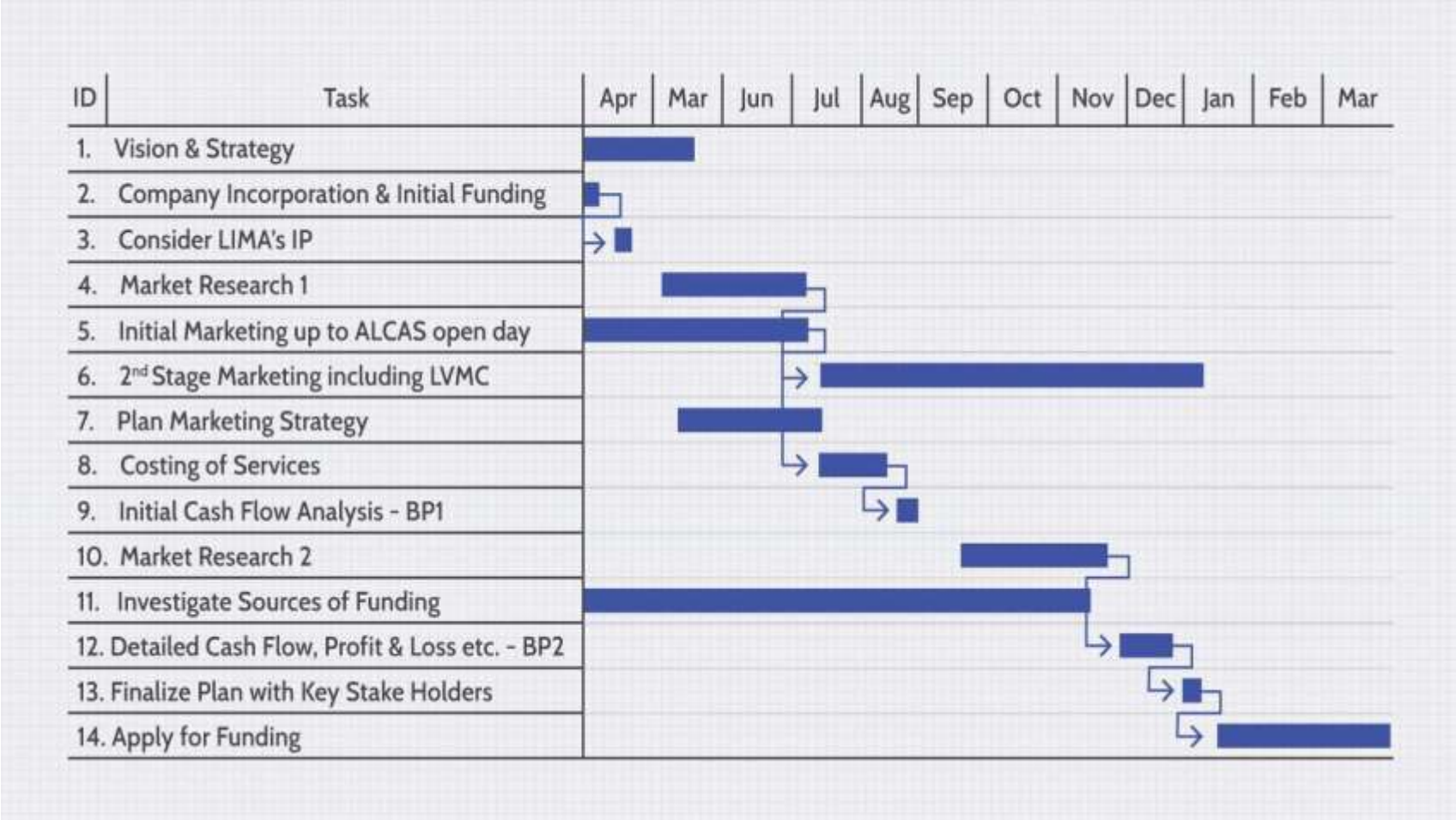
Project Scheduling

- A schedule is the conversion of a project WBS into an operating timetable.
- A properly designed detailed schedule can serve as a key input in establishing the monitoring and control system of the project.
- Project Milestone is an effective tool.

Milestone

- Also known as Mile marker.
- Milestone is used for significant delivery, end of a project phase and key activity in a project..
- Gantt Chart.

Gantt Chart


ID
Task
Apr
Mar
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Feb
Mar
1. Vision & Strategy
2. Company Incorporation & Initial Funding
3. Consider LIMA's IP
4. Market Research 1
5. Initial Marketing up to ALCAS open day
6. 2nd Stage Marketing including LVMC
7. Plan Marketing Strategy
8. Costing of Services
9. Initial Cash Flow Analysis - BP1
10. Market Research 2
11. Investigate Sources of Funding
12. Detailed Cash Flow, Profit & Loss etc. - BP2
13. Finalize Plan with Key Stake Holders
14. Apply for Funding

Project Controlling

Computerized programs produce a broad variety of PERT/CPM reports, including

(1) detailed cost breakdowns for each task, (2) total program labor curves, (3) cost distribution tables, (4) functional cost and hour summaries, (5) raw material and expenditure forecasts, (6) variance reports, (7) time analysis reports, and (8) work status reports.

Project Management Techniques

- Critical Path Method (CPM) and Program evaluation and review technique (PERT) are extremely valuable.

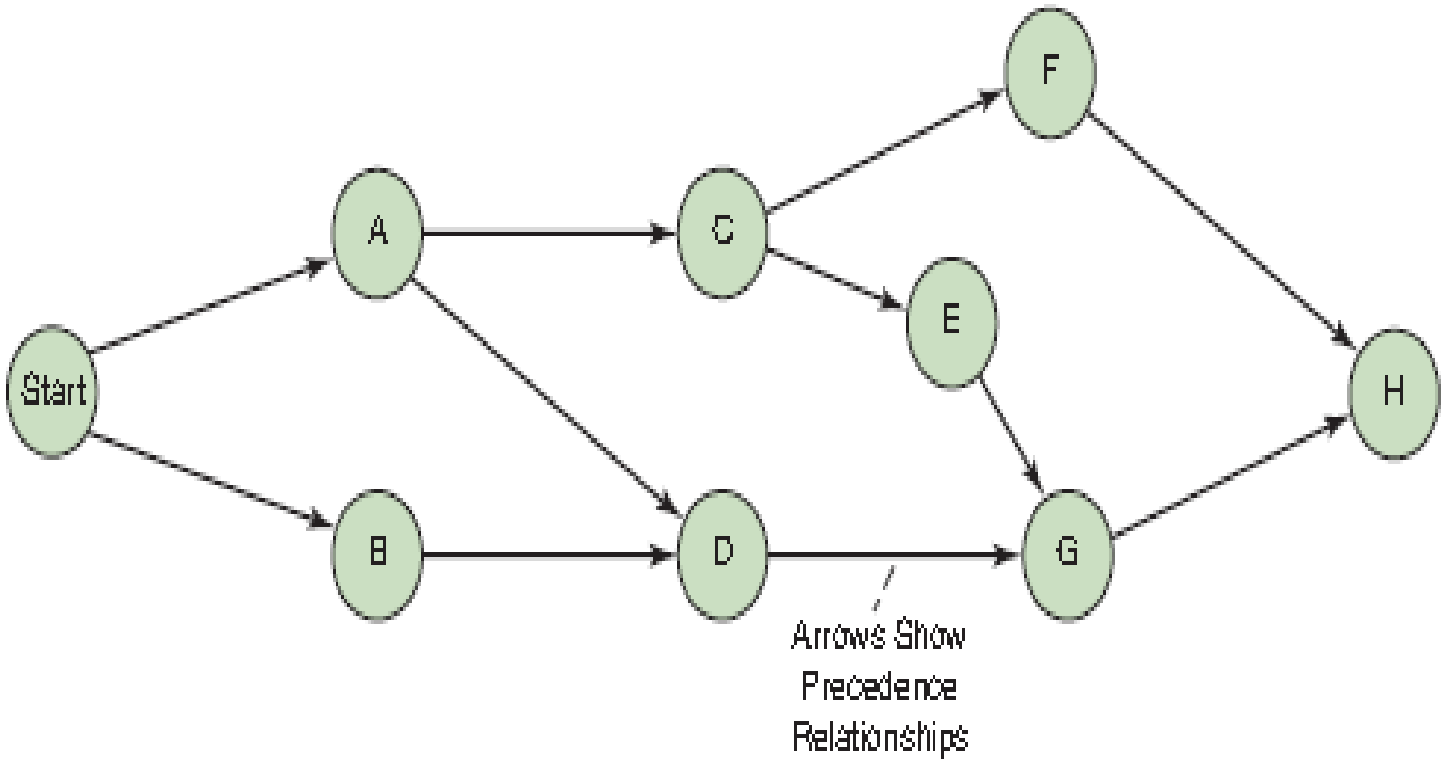
Project Management Techniques

- Program evaluation and review technique (PERT) : A project management technique that employs three time estimates for each activity.
- Critical path method (CPM) : A project management technique that uses only one time factor per activity.
- Critical path : The computed *longest* time path(s) through a network.

Network Diagram

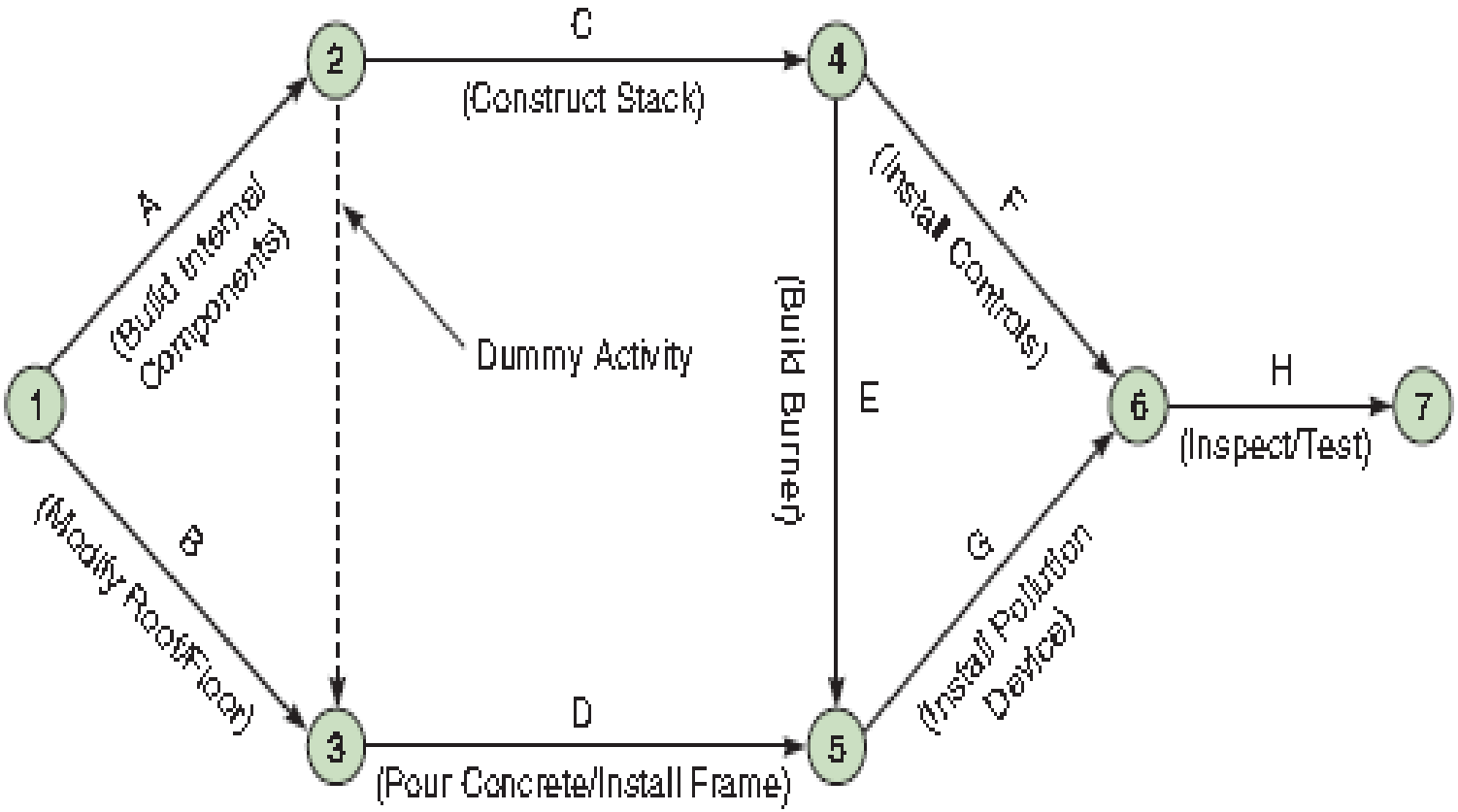
- Activity-on-node (AON) : A network diagram in which nodes designate activities.
- Activity-on-arrow (AOA) : A network diagram in which arrows designate activities.
- **Dummy activity** —An activity having no time that is inserted into a network to maintain the logic of the network.

AON Diagram



Source: Heizer, Render, Muckson

AOA Diagram



Source: Heizer, Render, Muckson

PDM

- The precedence diagramming method (PDM) is a technique used for constructing a schedule model in which activities are represented by nodes and are graphically linked by one or more logical relationships to show the sequence in which the activities are to be performed.
- The activity that must be completed before a particular activity can begin is known as immediate predecessor.
- PDM includes four types of dependencies or logical relationships.

PDM

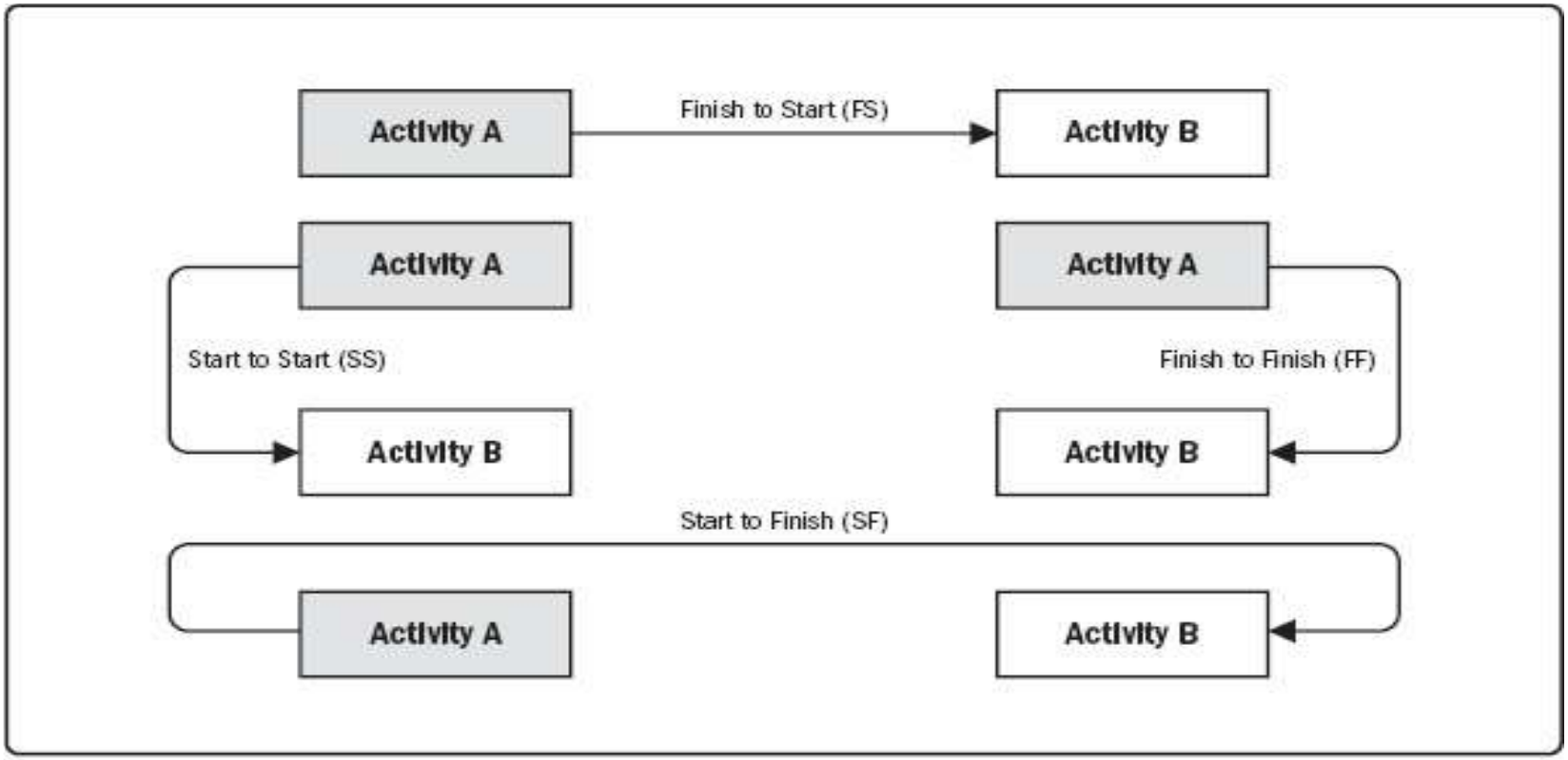


Figure 6-9. Precedence Diagramming Method (PDM) Relationship Types

Source : PMBOK

Project Schedule Network Diagram

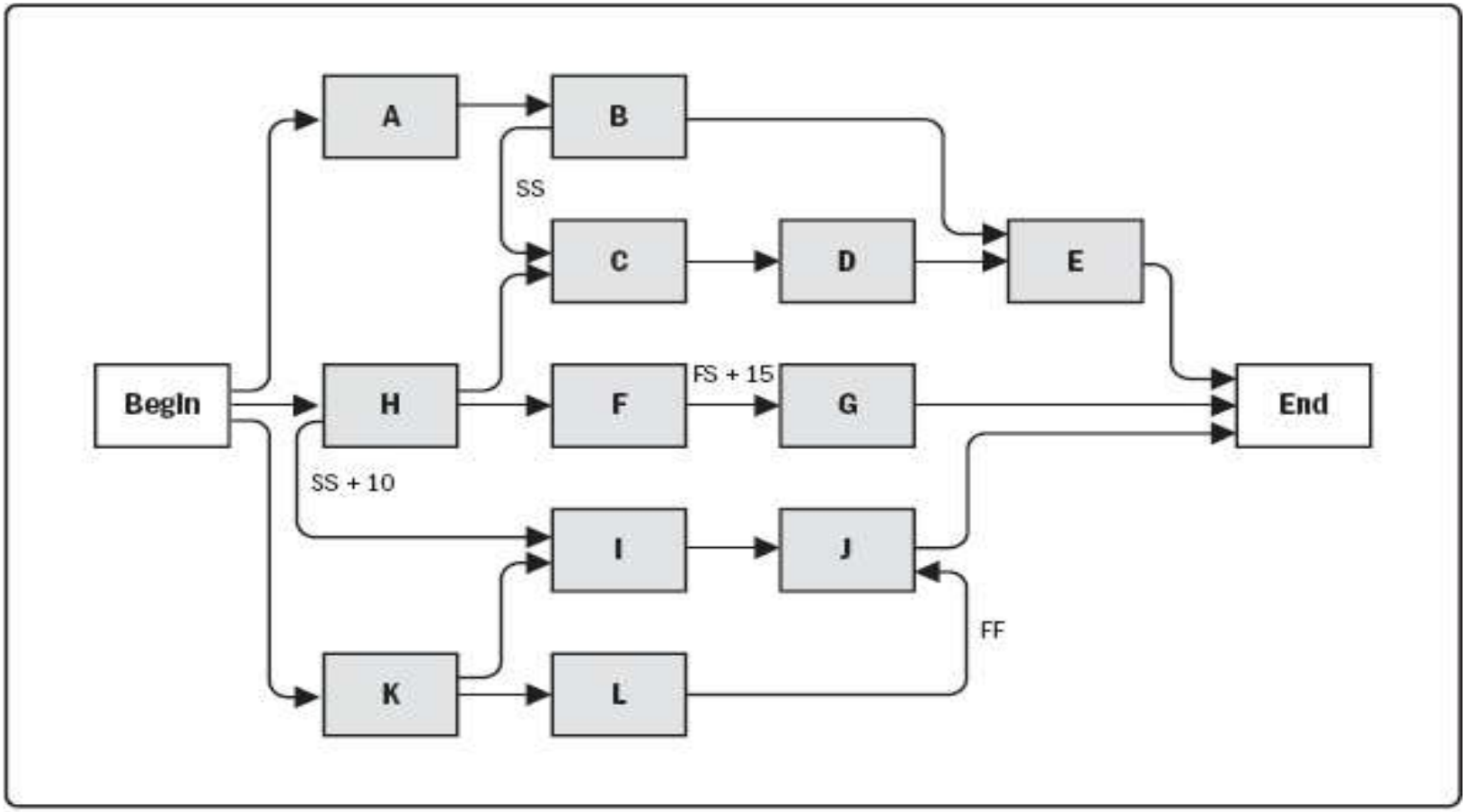


Figure 6-11. Project Schedule Network Diagram

Source : PMBOK

Critical Path Method (CPM)

- CPM used deterministic estimates and allowed project crunching.
- CPM offered the option of reducing activity times by adding more workers and/or resources at an increased cost.
- CPM can trade off between cost and time.
- Critical path is the path through the project where if any activity is delayed, the project will be delayed.

Key Terms

- Earliest start (ES) = earliest time at which an activity can start, assuming all predecessors have been completed.
- Earliest finish (EF) = earliest time at which an activity can be finished.
- Latest start (LS) = latest time at which an activity can start so as to not delay the completion time of the entire project.
- Latest finish (LF) = latest time by which an activity has to finish so as to not delay the completion time of the entire project.

Class Exercise

- The owner of a shopping center plans to modernize and expand the current 32-business shopping center complex.
- The project is expected to provide room for 8-10 new businesses.
- The finance has been arranged through venture capitalist.
- The goal is to schedule the project to know when it will be complete.

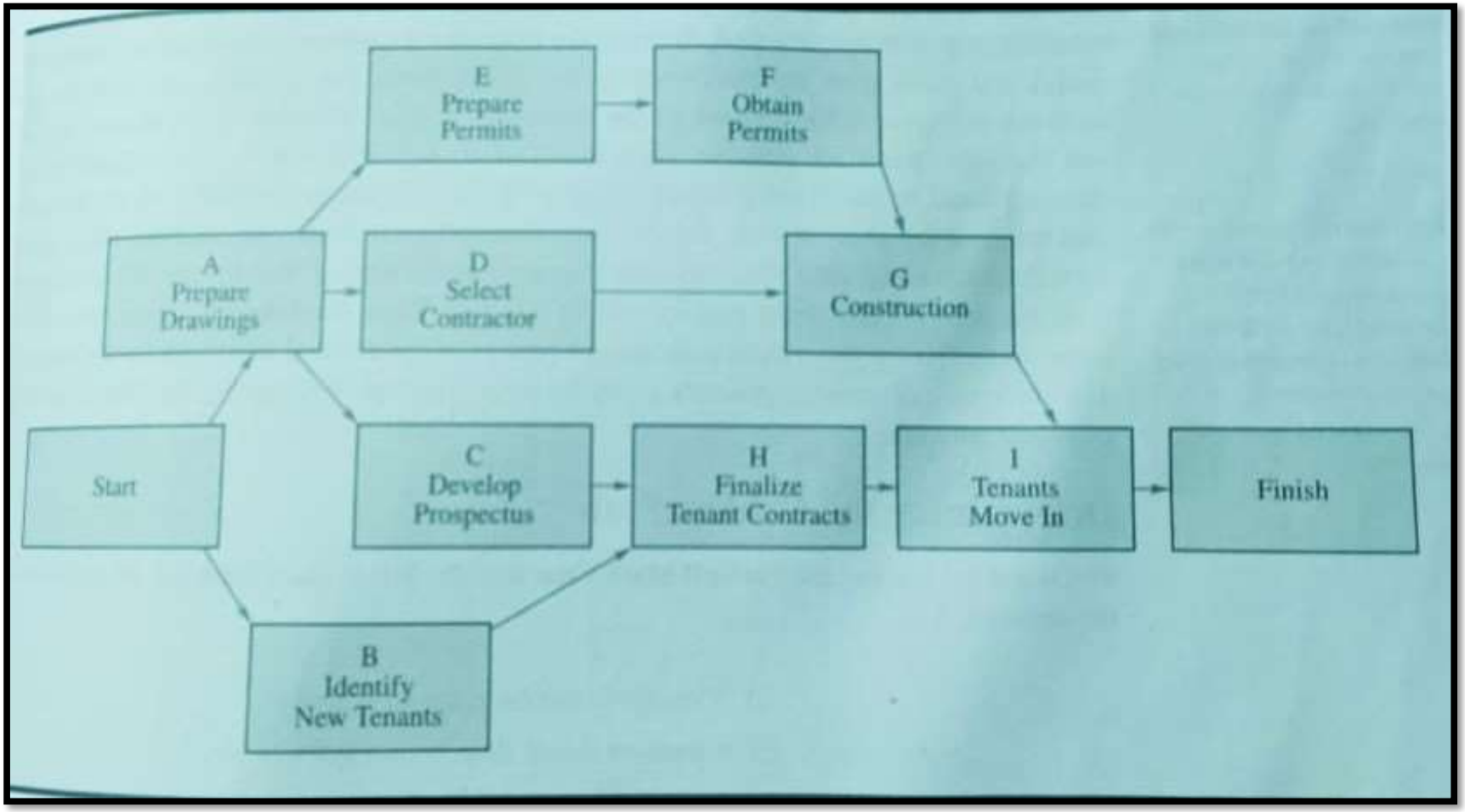
Step 1 : List of Activities

TABLE 9.1 LIST OF ACTIVITIES FOR THE WESTERN HILLS SHOPPING CENTER PROJECT

Activity	Activity Description	Immediate Predecessor	Expected Activity Time
A	Prepare architectural drawings	—	5
B	Identify potential new tenants	—	6
C	Develop prospectus for tenants	A	4
D	Select contractor	A	3
E	Prepare building permits	A	1
F	Obtain approval for building permits	E	4
G	Perform construction	D, F	14
H	Finalize contracts with tenants	B, C	12
I	Tenants move in	G, H	2
		Total	51

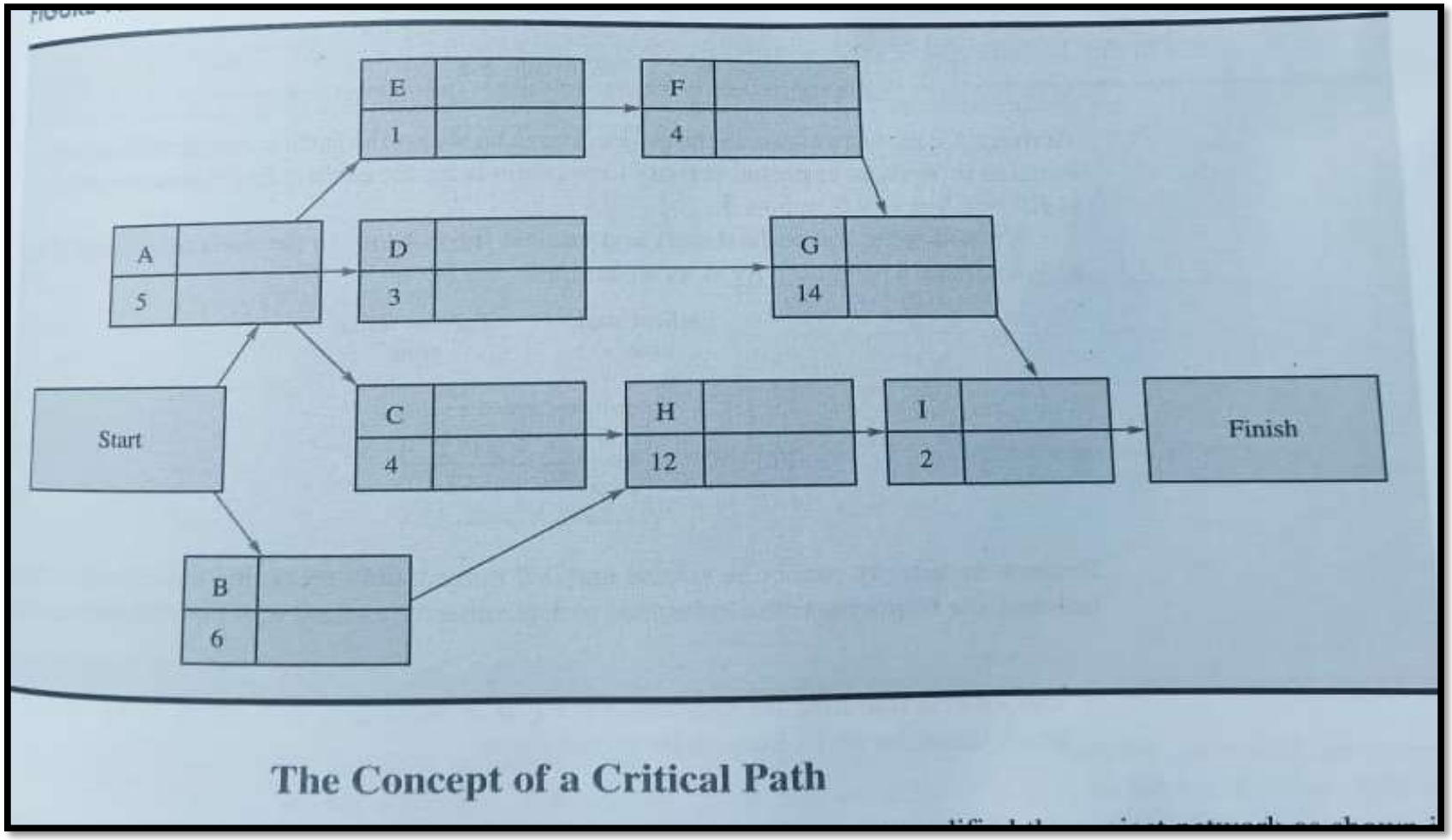
Source : An Introduction to Management Science

Step 2: Network Diagram



Source : An Introduction to Management Science

Step 3 : Draw Network Diagram with Time



Source : An Introduction to Management Science

Step 4 : Determination of ES and EF

Determining the Critical Path

We begin by finding the **earliest start time** and the **latest start time** for all activities in the network. Let

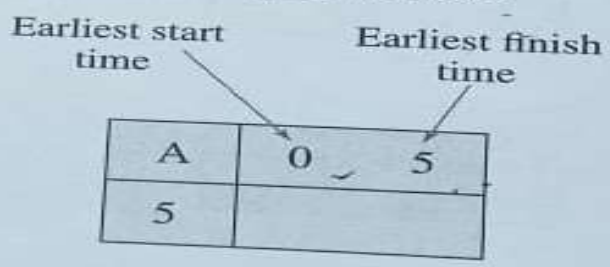
$$ES = \text{earliest start time for an activity}$$
$$EF = \text{earliest finish time for an activity}$$
$$t = \text{expected activity time}$$

The **earliest finish time** for any activity is

$$EF = ES + t \tag{9.1}$$

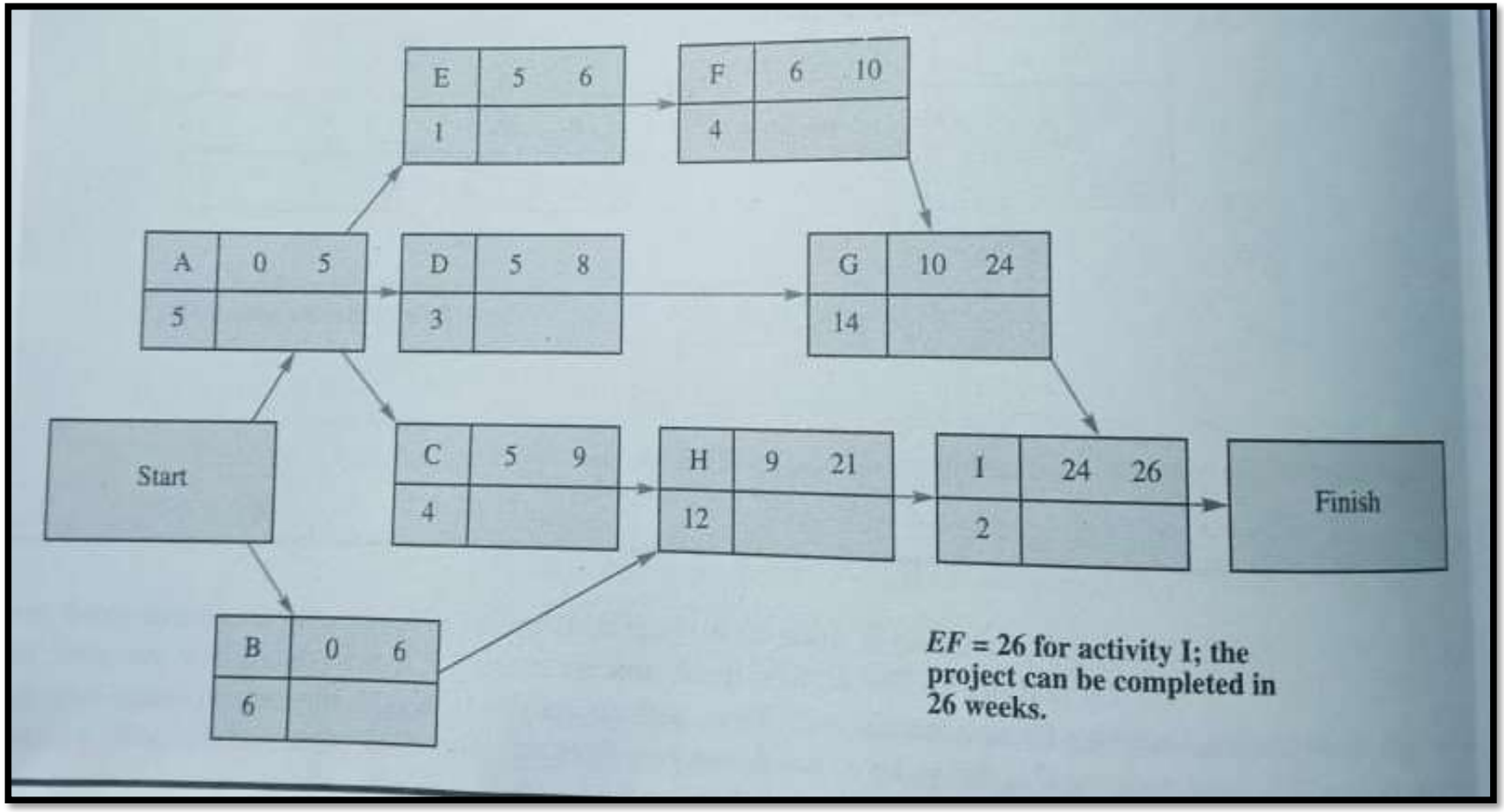
Activity A can start as soon as the project starts, so we set the earliest start time for activity A equal to 0. With an expected activity time of 5 weeks, the earliest finish time for activity A is $EF = ES + t = 0 + 5 = 5$.

We will write the earliest start and earliest finish times in the node to the right of the activity letter. Using activity A as an example, we have



Source : An Introduction to Management Science

Step 5 : Scheduling—Forward Pass



Source : An Introduction to Management Science

Step 6 : Determination of LS and LF

LS = latest start time for an activity

LF = latest finish time for an activity

Then

$$LS = LF - t \tag{9.2}$$

$LS = LF - t = 26 - 2 = 24$. We will write the LS and LF values in the node directly below the earliest start (ES) and earliest finish (EF) times. Thus, for node I, we have

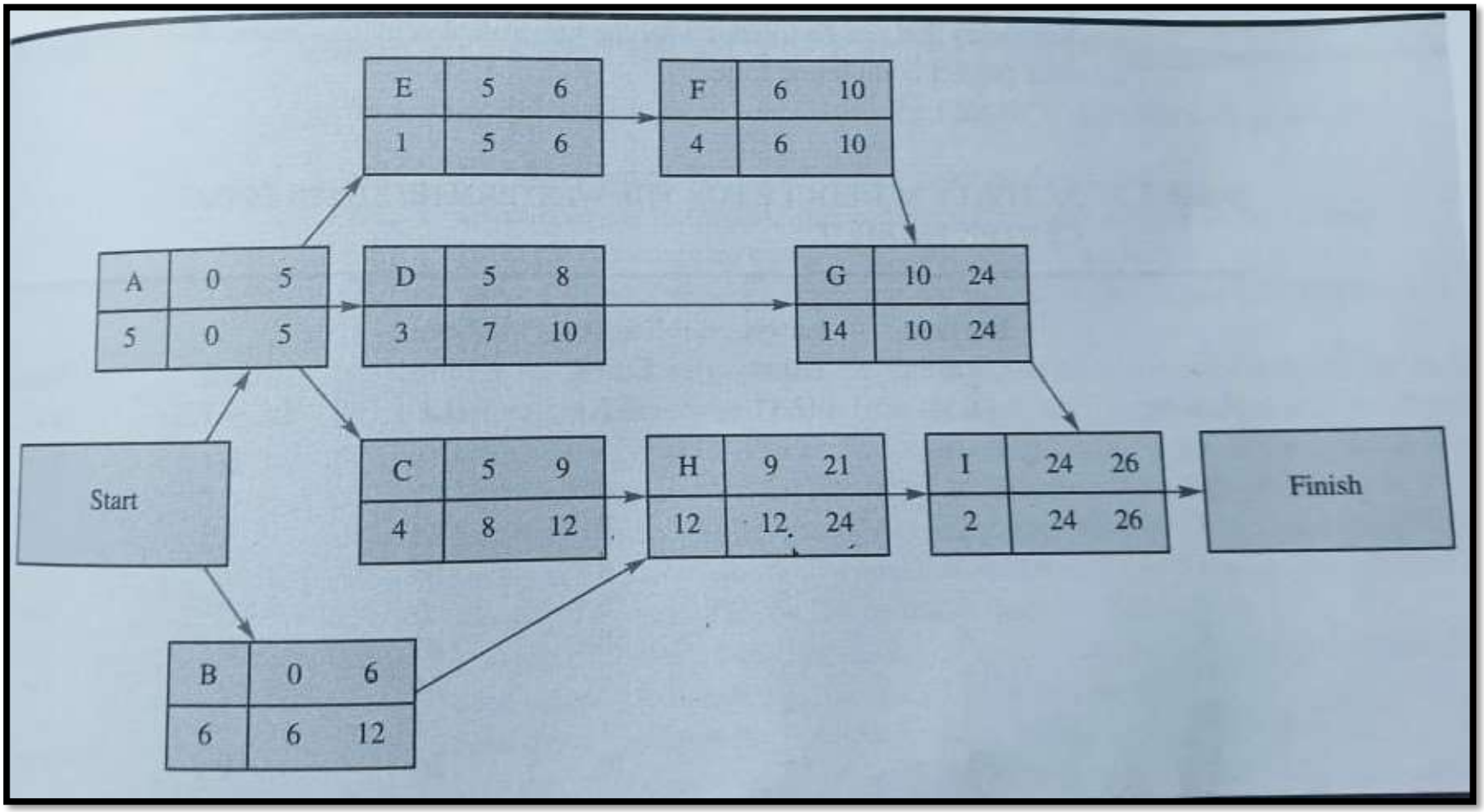
I	24	26
2	24	26

Latest start time

Latest finish time

Source : An Introduction to Management Science

Step 7 : Scheduling Backward Pass



Source : An Introduction to Management Science

Step 8: Slack and Activity Schedule

$$\text{Slack} = LS - ES = LF - EF$$

For example, the slack associated with activity C is 2.

TABLE 9.2 ACTIVITY SCHEDULE FOR THE WESTERN HILLS SHOPPING CENTER PROJECT

Activity	Earliest Start (ES)	Latest Start (LS)	Earliest Finish (EF)	Latest Finish (LF)	Slack ($LS - ES$)	Critical Path?
A	0	0	5	5	0	Yes
B	0	6	6	12	6	
C	5	8	9	12	3	
D	5	7	8	10	2	
E	5	5	6	6	0	Yes
F	6	6	10	10	0	Yes
G	10	10	24	24	0	Yes
H	9	12	21	24	3	
I	24	24	26	26	0	Yes

Source : An Introduction to Management Science

Step 9: Critical Path

A-E-F-G-I

Any Question

PERT

- To incorporate uncertain activity times into the analysis, we need to obtain three-time estimates for each activity (probabilistic time estimates).
- Optimistic time (a) = the minimum activity time if everything progress ideally.
- Most probable time (m) = The most probable activity time under normal conditions.
- Pessimistic time (b) = The maximum activity time if substantial delays are encountered.

$$\textbf{Expected time (t)} = \frac{a+4m+b}{6}$$

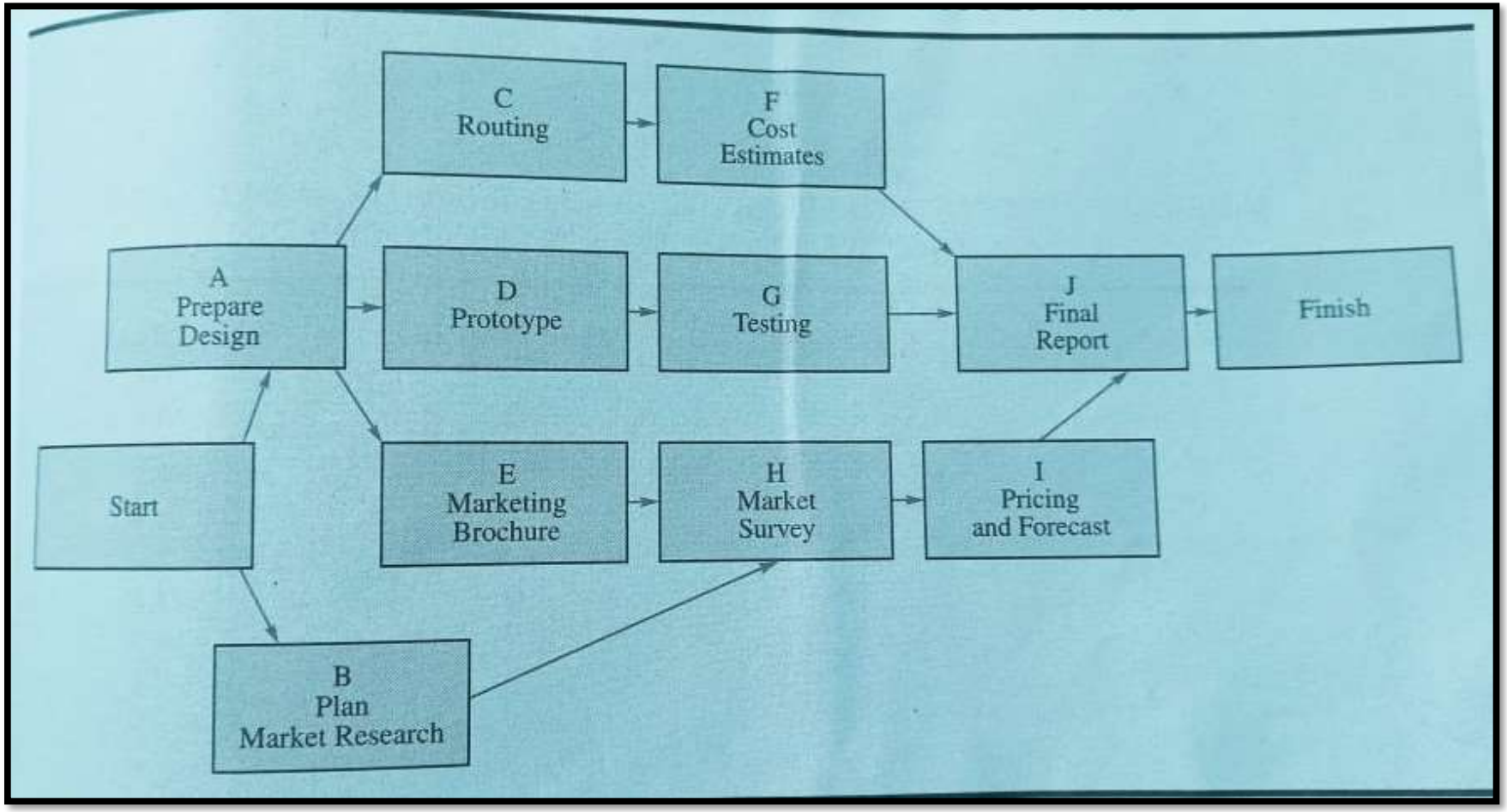
$$\textbf{Variance} = \left(\frac{b-a}{6}\right)^2$$

Step 1: Activity List

Activity	Description	Immediate Predecessor
A	Develop product design	—
B	Plan market research	—
C	Prepare routing (manufacturing engineering)	A
D	Build prototype model	A
E	Prepare marketing brochure	A
F	Prepare cost estimates (industrial engineering)	C
G	Do preliminary product testing	D
H	Complete market survey	B, E
I	Prepare pricing and forecast report	H
J	Prepare final report	F, G, I

Source : An Introduction to Management Science

Step 2: Draw Network Diagram



Source : An Introduction to Management Science

Step 3: Activity Time

TABLE 9.4 OPTIMISTIC, MOST PROBABLE, AND PESSIMISTIC ACTIVITY TIME ESTIMATES (IN WEEKS) FOR THE PORTA-VAC PROJECT

Activity	Optimistic (a)	Most Probable (m)	Pessimistic (b)
A	4	5	12
B	1	1.5	5
C	2	3	4
D	3	4	11
E	2	3	4
F	1.5	2	2.5
G	1.5	3	4.5
H	2.5	3.5	7.5
I	1.5	2	2.5
J	1	2	3

Source : An Introduction to Management Science

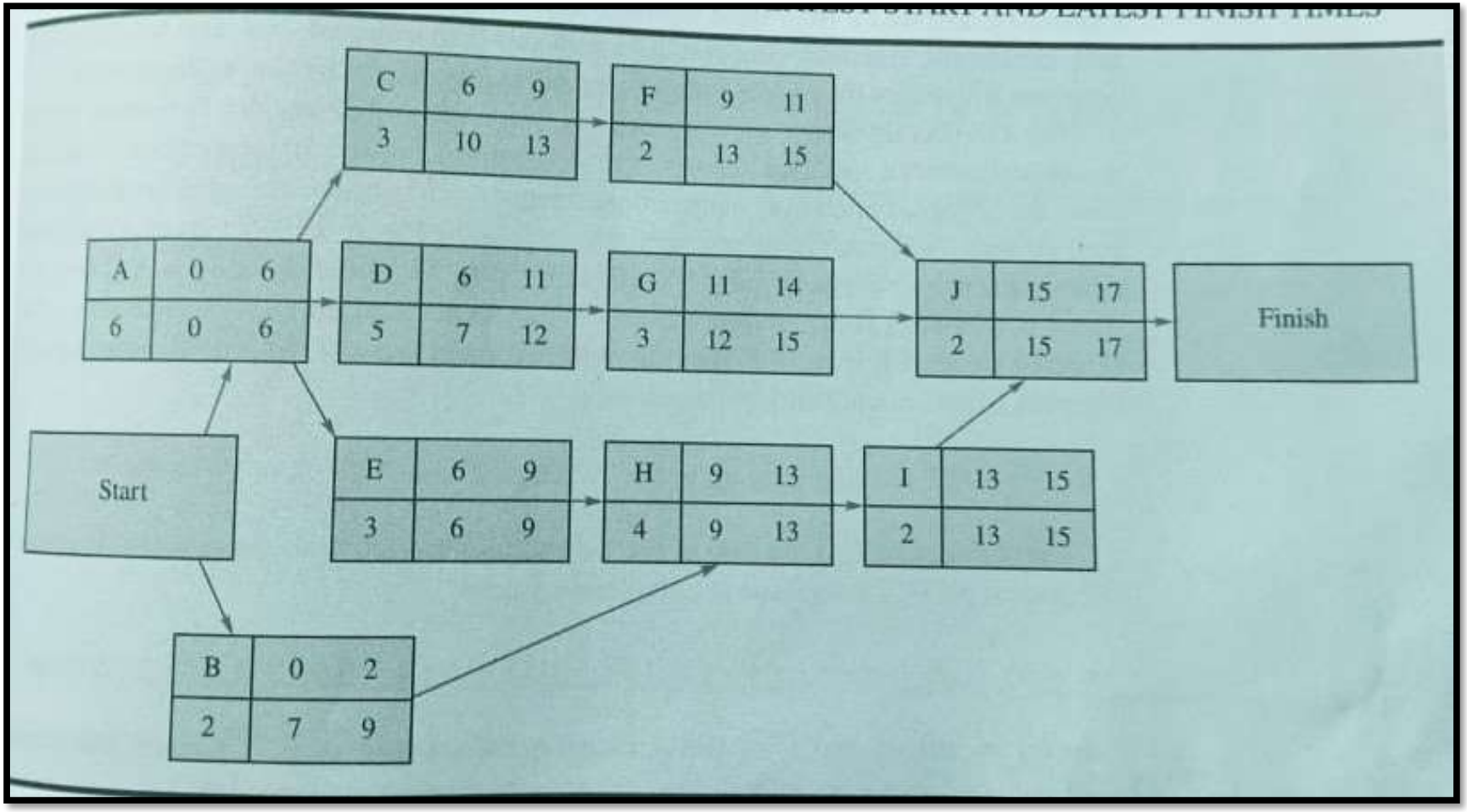
Step 4: Determine Expected Time and Variance

TABLE 9.5 EXPECTED TIMES AND VARIANCES FOR THE PORTA-VAC PROJECT ACTIVITIES

Activity	Expected Time (weeks)	Variance
A	6	1.78
B	2	0.44
C	3	0.11
D	5	1.78
E	3	0.11
F	2	0.03
G	3	0.25
H	4	0.69
I	2	0.03
J	2	0.11
Total	32	

Source : An Introduction to Management Science

Step 7: Scheduling Forward and Backward Pass



Source : An Introduction to Management Science

Step 8: Slack and Activity Schedule

Activity	Earliest Start (*ES*)	Latest Start (*LS*)	Earliest Finish (*EF*)	Latest Finish (*LF*)	Slack (*LS* – *ES*)	Critical Path?
A	0	0	6	6	0	Yes
B	0	7	2	9	7	
C	6	10	9	13	4	
D	6	7	11	12	1	
E	6	6	9	9	0	Yes
F	9	13	11	15	4	
G	11	12	14	15	1	
H	9	9	13	13	0	Yes
I	13	13	15	15	0	Yes
J	15	15	17	17	0	Yes

Source : An Introduction to Management Science

Step 9: PERT Analysis

$$E(T_1) = t_A + t_E + t_H + t_I + t_J = 6 + 3 + 4 + 2 + 2 = 17 \text{ weeks}$$

The variance of T_i is the sum of the variances of the activities along path i. For path 1 (the critical path), the variance in completion time is

$$\sigma_1^2 = \sigma_A^2 + \sigma_E^2 + \sigma_H^2 + \sigma_I^2 + \sigma_J^2 = 1.78 + 0.11 + 0.69 + 0.03 + 0.11 = 2.72 \text{ weeks}^2$$

Source : An Introduction to Management Science

Advantages of PERT/CPM

1. Especially useful when scheduling and controlling large projects
2. Straightforward concept and not mathematically complex
3. Graphical networks help highlight relationships among project activities
4. Critical path and slack time analyses help pinpoint activities that need to be closely watched

Advantages of PERT/CPM

5. Project documentation and graphics point out who is responsible for various activities
6. Applicable to a wide variety of projects
7. Useful in monitoring not only schedules but costs as well

Limitations of PERT/CPM

1. Project activities have to be clearly defined, independent, and stable in their relationships
2. Precedence relationships must be specified and networked together
3. Time estimates tend to be subjective and are subject to fudging by managers
4. There is an inherent danger of too much emphasis being placed on the longest, or critical, path

Any Question

Probability

PROBABILITY OF COMPLETING A PROJECT ON TIME

Julie Ann Williams would like to find the probability that her project will be finished on or before the 16-week Earth Day deadline.

APPROACH ▶ To do so, she needs to determine the appropriate area under the normal curve. This is the area to the left of the 16th week.

SOLUTION ▶ The standard normal equation can be applied as follows:

$$Z = (\text{Due date} - \text{Expected date of completion})/\sigma_p$$
$$= (16 \text{ weeks} - 15 \text{ weeks})/1.76 \text{ weeks} = 0.57 \qquad (3\text{-}9)$$

Probability is 71.57 %

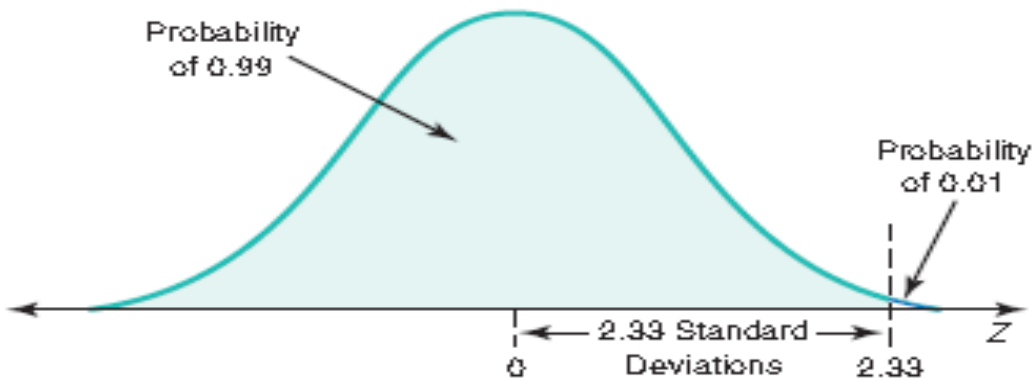
Source: Heizer, Render, Muckson

Probability

COMPUTING PROBABILITY FOR ANY COMPLETION DATE

Julie Ann Williams wants to find the due date that gives her company's project a 99% chance of *on-time* completion.

APPROACH ► She first needs to compute the Z-value corresponding to 99%, as shown in Figure 3.14. Mathematically, this is similar to Example 10, except the unknown is now the due date rather than Z.



SOLUTION ► Referring again to the Normal Table in Appendix I (alternatively using the Excel formula =NORMSINV(0.99)), we identify a Z-value of 2.33 as being closest to the probability of 0.99. That is, Julie Ann Williams's due date should be 2.33 standard deviations above the mean project completion time. Starting with the standard normal equation [see Equation (3-9)], we can solve for the due date and rewrite the equation as:

$$\begin{aligned}\text{Due date} &= \text{Expected completion time} + (Z \times \sigma_p)\\ &= 15 + (2.33 \times 1.76) = 19.1 \text{ weeks} \qquad (3\text{-}10)\end{aligned}$$

Source: Heizer, Render, Muckson

Any Question

Schedule Compression

Crashing.

A technique used to shorten the schedule duration for the least incremental cost by adding resources.

Fast tracking.

A schedule compression technique in which activities or phases normally done in sequence are performed in parallel for at least a portion of their duration.

Schedule Compression

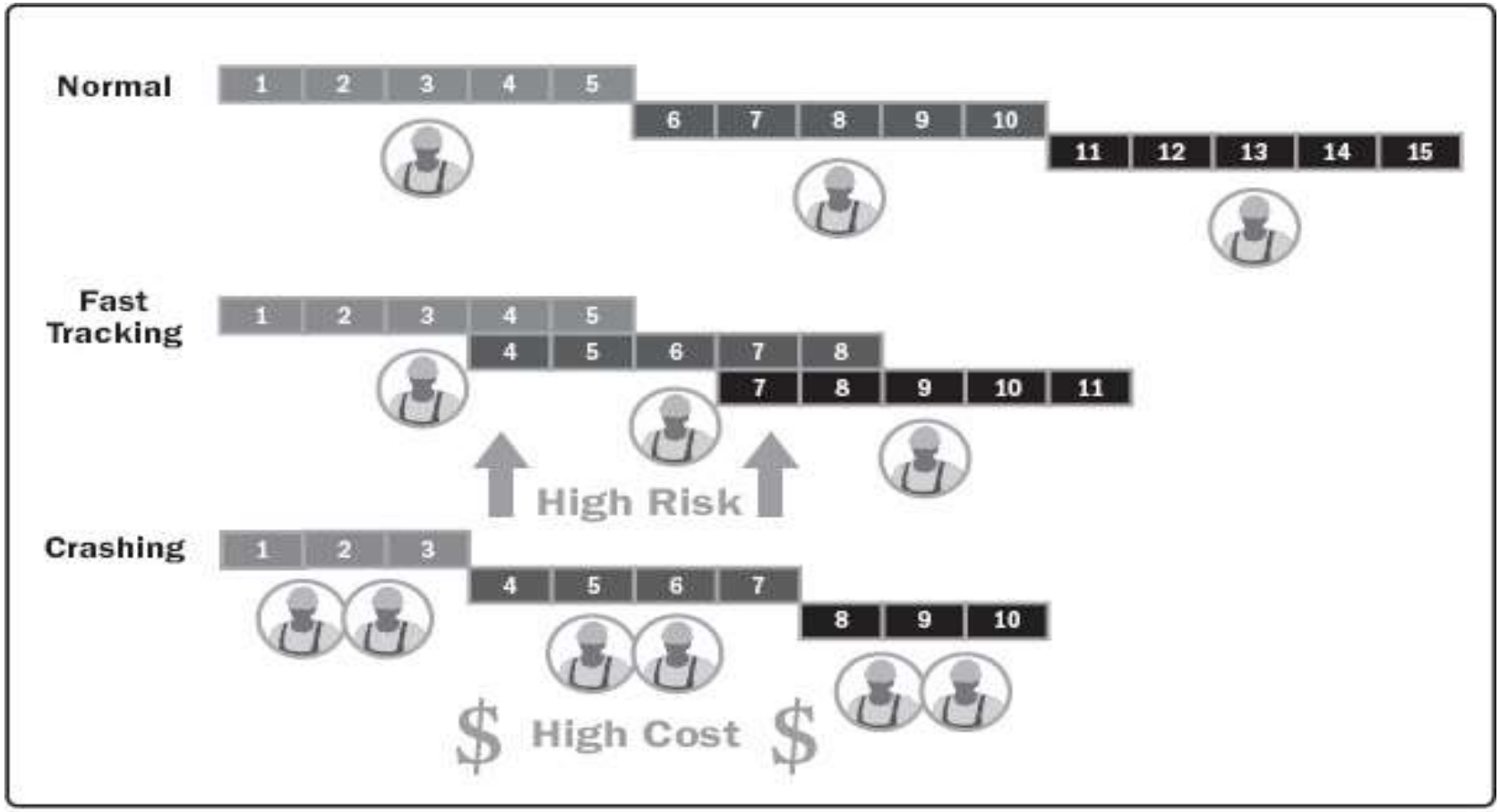


Figure 6-19. Schedule Compression Comparison

Source: PMBOK

Slack and Activity Schedule : CPM

$$\text{Slack} = LS - ES = LF - EF$$

For example, the slack associated with activity C is 2.

TABLE 9.2 ACTIVITY SCHEDULE FOR THE WESTERN HILLS SHOPPING CENTER PROJECT

Activity	Earliest Start (ES)	Latest Start (LS)	Earliest Finish (EF)	Latest Finish (LF)	Slack ($LS - ES$)	Critical Path?
A	0	0	5	5	0	Yes
B	0	6	6	12	6	
C	5	8	9	12	3	
D	5	7	8	10	2	
E	5	5	6	6	0	Yes
F	6	6	10	10	0	Yes
G	10	10	24	24	0	Yes
H	9	12	21	24	3	
I	24	24	26	26	0	Yes

Source : An Introduction to Management Science

Slack and Activity Schedule :PERT

Activity	Earliest Start (*ES*)	Latest Start (*LS*)	Earliest Finish (*EF*)	Latest Finish (*LF*)	Slack (*LS* − *ES*)	Critical Path?
A	0	0	6	6	0	Yes
B	0	7	2	9	7	
C	6	10	9	13	4	
D	6	7	11	12	1	
E	6	6	9	9	0	Yes
F	9	13	11	15	4	
G	11	12	14	15	1	
H	9	9	13	13	0	Yes
I	13	13	15	15	0	Yes
J	15	15	17	17	0	Yes

Source : An Introduction to Management Science

Project Crashing

Crashing a project involves four steps:

STEP 1: Compute the crash cost per week (or other time period) for each activity in the network. If crash costs are linear over time, the following formula can be used:

$$\text{Crash cost per period} = \frac{(\text{Crash cost} - \text{Normal cost})}{(\text{Normal time} - \text{Crash time})} \qquad (3\text{-}11)$$

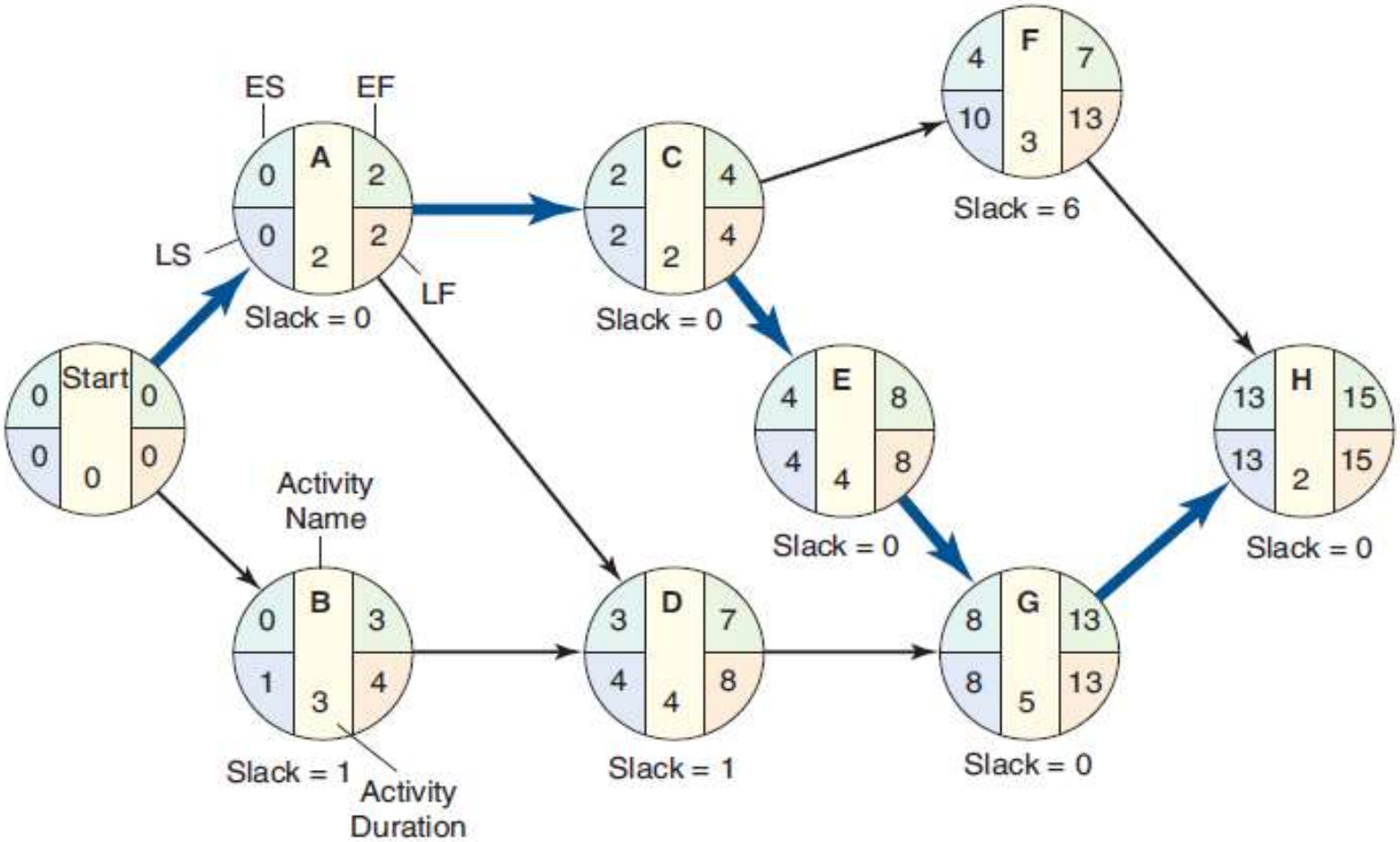
STEP 2: Using the current activity times, find the critical path(s) in the project network. Identify the critical activities.

STEP 3: If there is only one critical path, then select the activity on this critical path that (a) can still be crashed and (b) has the smallest crash cost per period. Crash this activity by one period.

If there is more than one critical path, then select one activity from each critical path such that (a) each selected activity can still be crashed and (b) the total crash cost per period of *all* selected activities is the smallest. Crash each activity by one period. Note that the same activity may be common to more than one critical path.

STEP 4: Update all activity times. If the desired due date has been reached, stop. If not, return to Step 2.

Source: Heizer, Render, Muckson



Source: Heizer, Render, Muckson

TABLE 3.5 Normal and Crash Data for Milwaukee Paper Manufacturing

ACTIVITY	TIME (WEEKS)		COST ($)		CRASH COST PER WEEK ($)	CRITICAL PATH?
	NORMAL	CRASH	NORMAL	CRASH		
A	2	1	22,000	22,750	750	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	3	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	84,500	1,500	Yes
H	2	1	16,000	19,000	3,000	Yes

Source: Heizer, Render, Muckson

Take Home Exercise

- Example 1-12 of your Text Book

Any Question

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 3
2. A Guide to the Project Management Body of Knowledge (PMBOK® Guide)* by Project Management Institute, published by Project Management Institute.
3. Project Management: A Managerial Approach Book by Jack R. Meredith, Samuel J. Mantel, and Scott M. Shafer.
4. An Introduction to Management Science by Anderson and Williams.